CHAPTER 5

Vehicle Operating Cost (VOC) Report

Annual Maintenance Plan F.Y. 2021-22

Table of Contents

1 Introduction
2 Results of Economic Analysis
3 Components of VOC

List of Tables

Table - 1: Selected Vehicle Type and Prices
Table - 2: Breakdown of Vehicle Cost
Table - 3: POL Cost in 2022
Table - 4: Market/Economic Costs of Motorcycles & Cars
Table - 5: Market/Economic Costs of Jeep, Pickup & Wagon
Table - 6: Market/Economic Costs of Bus & Trucks
Table - 7: Total Vehicle Operating Cos (Economic) Per 1000 Vehicle – Km. in Rupees
Table - 8: Total Vehicle Operating Cos (Economic) Per 1000 Vehicle – Km. in Rupees
Table - 9: Total Vehicle Operating Cos (Economic) Per 1000 Vehicle – Km. in Rupees
Table - 10: Percentage Distribution of VOC Components

List of Figures

Figure - 1: VOC per 1000 – Km of Different Vehicles
Figure - 2: Present the Distribution of Components of VOC of Different Vehicle Types

1. INTRODUCTION

This report computes vehicle operating costs based on Free Flow Vehicle Operating Cost Model Version 4.0 (**HDM-VOC**) of the World Bank. For analysis current market prices of vehicle, tires and Fuel & Lubricants etc. have been collected during January - February, 2022.

Typical market prices of vehicles are described at Table-1.

Table - 1 Selected Vehicle Types and Market Prices

Vehicle Type, Make & Model	Engine Power (cc)	Factory Gate Price (Rupees)	Remarks
Motor Cycles			
Yamaha 125 Z	125	151000	Complete Knock Down
Honda 125	125	152000	CKD
Honda 70	70	95000	CKD
CAR			
Toyota Corolla Yaris (standard)	1296	2,800,000	CKD
Toyota Corolla Altis 1.3 (standard)	1299	3,350,000	CKD
Toyota Corolla Altis 1.8 (standard)	1800	3,833,500	CKD
Suzuki Mehran (standard)	796	860,000	CKD
Suzuki Cultus (standard)	1300	2,244,000	CKD
Honda City	1200	3,129,000	CKD
JEEP			
Toyota Fortuner 2.7 V	2694	9,399,000	Complete Built Unit
PICKUP AND UTILITY			
Toyota Hilux (Single) 4x2	2446	2,876,500	2.58 tonne, CKD
Toyota Hilux (Grade E)	2779	4,127,000	2.58 tonne, CKD
Hyundai SHEHZORE	2600	1,889,000	2.73 tonne, CKD
WAGON AND MINIBUS			
Toyota Hiace (standard)	2800	7,399,000	16 seats, CBU
Toyota Coaster	3661	14,809,000	26 seats, CBU
Hino FB2WGKZ	4000	5,900,000	29 seats, CKD
BUS			
Hino AK 3HRKA (46 seater)	7412	11,800,000	54 seats CKD
TRUCK 2-Axle with body			
Hino FL	7961	10,511,000	CKD
TRUCK 3-Axle with body			
Hino	7961	13,500,000	CKD
TRUCK TRACTOR with trailer			
Hino SG	10900	12,700,000	CKD

The market prices of vehicles, tires and POL have been converted into economic costs by deducting taxes, duties, transfer payment etc. Table –2 shows the tariff structure as applicable on these items.

Table - 2 Breakdown of Vehicle Cost

Vehicle Type & Cost	Charges Detail	Formula	Rate	Remarks
Motor Cycle, Car and Other Small Vehicles (CKD)				
a	Cost, Insurance and Freight		100%	CBU stands for Complete built-up Unit
b	Custom Duty		30%	
c	Port Charges		5%	
d	Sub-Total	(a+b+c)	135%	CKD stands for Complete Knock Down
e	Mark-ups	(20% of d)	27%	
f	Sub-Total	(d+e)	162%	
g	Sales Tax	(15% of f)	24%	
h	Factory Gate Price	(f+g)	186%	
i	ECONOMIC COST	(a+c+e) / h	**71%**	
j	Taxes	(100 - I)	**29%**	
WAGONS, MINI-BUS (16 SEATS) (CBU)				
a	Cost, Insurance and Freight		100%	
b	Custom Duty		60%	
c	Port Charges		5%	
d	Sub-Total	(a+b+c)	165%	
e	Mark-ups	(5% of d)	8%	
f	Sub-Total	(d+e)	173%	
g	Sales Tax	(15% of f)	26%	
h	Factory Gate Price	(f+g)	199%	
i	ECONOMIC COST	(a+c+e) / h	**57%**	
j	Taxes	(100 - I)	**43%**	
BUSES & TRUCKS (CKD)				
a	Cost, Insurance and Freight		100%	
b	Custom Duty		30%	
c	Port Charges		5%	
d	Sub-Total	(a+b+c)	135%	
e	Mark-ups	(40% of d)	54%	
f	Sub-Total	(d+e)	189%	
g	Sales Tax	(15% of f)	28%	
h	Factory Gate Price	(f+g)	217%	
I	ECONOMIC COST	(a+c+e) / h	**73%**	
J	Taxes	(100 - I)	**27%**	
ALL VEHICLES USING LOCAL PRODUCTS				
a	Cost and Mark-ups		100%	
b	Sales Tax		15%	
c	Factory Gate Price		115%	
d	ECONOMIC COST	(a / c)	**87%**	
e	Taxes	(100 - I)	**13%**	
USING IMPORTED TYRES				
a	Cost, Insurance and Freight		100%	
b	Custom Duty		40%	
c	Port Charges		5%	

d	Sub-Total	(a+b+c)	145%	
e	Mark-ups	(5% of d)	7%	
f	Sub-Total	(d+e)	152%	
g	Sales Tax	(15% of f)	23%	
h	Factory Gate Price	(f+g)	175%	
i	ECONOMIC COST	(a+c+e) / h	**64%**	
j	Taxes	(100 - I)	**36%**	

Table – 3
POL Cost in 2022

			FINANCIAL		ECONOMIC	
	Petrol	Diesel	Petrol	Diesel	Petrol	Diesel
Ex-Refinery	88.7	89.0	88.7	89.0	88.7	89.0
Excise Duty (0% of No.1)	0.0	0.0	0.0	0.0		
Petroleum Development Levy	34.1	29.4	34.1	29.4		
(No. 1+2+3)	122.8	118.4	122.8	118.4		
Freight Charges	2.1	1.7	2.1	1.7	2.1	1.7
(No. 4+5)	124.9	120.1	124.9	120.1		
Distributor's Margin	2.1	2.0	2.1	2.0	2.1	2.0
Dealer's Margin	2.6	2.2	2.6	2.2	2.6	2.2
Price Before Sale (6+7+8)	129.5	124.3	129.5	124.3		
GST (16%)	20.7	19.9	20.7	19.9		
Max. Ex Depot Sale Price (9+10)	150.3	144.2	150.3	144.2	95.5	95.0

As of March, 2022

Passenger Time Cost & Cost per Vehicle

Description	Vehicle				
	Motor Cycle	Car	Wagon	Mini-bus	Bus
Number of Occupants					
Prime Occupants	1	1	2	3	3
Valued Occupants	1	1	8	15	35
Total	2	2	10	18	38
Hours					
Working month (25 days * 8 hours)	200	200	200	200	200
Work related Trips @ 20%	40	40	40	40	40
Income (Rs.)					
Prime Occupants (per month)	25000	45000	0	0	0
Valued Occupants (per hour)	125	225	50	50	50
Time Cost per vehicle	62.5	112.5	199.92	374.85	874.65
Time Cost per passenger	62.5	112.5	24.99	24.99	24.99

Vehicle	Crew	Cost per vehicle (Rs.)

	Type	Number	Monthly	Hourly
Wagon	Driver	1	17000	
	Conductor	1	10200	
	Sub-Total	2	27200	136
Mini-bus				
	Driver	1	17000	
	Conductor	1	10200	
	Cleaner	1	8500	
	Sub-Total	3	35700	178.5
Bus				
	Driver	1	25500	
	Conductor	1	11900	
	Cleaner	1	8500	
	Sub-Total	3	45900	229.5
Truck Rigid				
	Driver	2	25500	
	Cleaner	1	8500	
	Sub-Total	4	34000	170
Truck Multi-Axle				
	Driver	2	25500	
	Cleaner	2	10200	
	Sub-Total	4	35700	178.5
Labour - All Vehicles	Mechanic	1	17000	100

By using the factors mentioned in Tables 2 & 3 the market prices of vehicles (CBU or CKD), POL, parts and tires have been converted into Economic Cost as detailed in Tables 4 to 6.

Table-4 Market / Economic Costs of Motorcycles & Cars

DESCRIPTION	Vehicle Types and Representative Model								
	Motorcycle			Car					
	Yamaha 125	Honda 70	Honda 125	Toyota Corolla Yaris	Toyota Corolla Altis 1.3	Toyota Corolla Altis 1.8	Suzuki Mehran (standard)	Suzuki Cultus (standard)	Honda City
	CKD								
BASIC SPECIFICATIONS									
Engine (cc)	125	70	125	1296	1299	1800	796	1300	1200
Gross Vehicle Weight (kg)	85	80	90	1730	1730	1850	620	800	1720
Number of Axles	-	-	-	2	2	2	2	2	2
Types of Tyres	270 - R18-4PR	270 - R18-4PR	270 - R18-4PR	175-70-R14	175-70-R14	185-65-R14	145-80-R12	165-80-R13	165-80-R13
Seats	2	2	2	5	5	5	5	5	5
USAGE									
Life (years)	10	10	10	12	12	12	12	12	12
Hours Driven per year	300	300	300	450	450	550	400	400	450
Average Speed (km/hr)	40	40	40	50	50	50	50	50	50
Km Driven per year	12000	12000	12000	22500	22500	27500	20000	20000	22500
COST (Rs)									
Vehicle Market Price	151,000	152,000	95,000	2,800,000	3,350,000	3,833,500	860,000	2,244,000	3,129,000
Market Price of Tyre	2,000	2,000	2,000	7,500	7,500	7,500	5,500	7,000	7,500
Market Price of Tyres in vehicle	4,000	4,000	4,000	30,000	30,000	30,000	22,000	28,000	30,000
-	-	-	-	-	-	-	-	-	-
Duty / Tax Ratio	0.29	0.29	0.29	0.29	0.29	0.29	0.29	0.29	0.29
Economic Cost Ratio	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71
Economic Cost	107,210	107,920	67,450	1,988,000	2,378,500	2,721,785	610,600	1,593,240	2,221,590
Tyre Cost (Rs)									
Market Price of Tyre	2,000	2,000	2,000	7,500	7,500	7,500	5,500	7,000	7,500
Tax Ratio on Tyre	0.13	0.13	0.13	0.13	0.36	0.36	0.13	0.13	0.13
Economic Cost Ratio	0.87	0.87	0.87	0.87	0.64	0.64	0.87	0.87	0.87
Economic Cost per Tyre	1,740	1,740	1,740	6,525	4,800	4,800	4,785	6,090	6,525

Table – 5 Market / Economic Costs of Jeeps, Pickup & Wagon

DESCRIPTION	Vehicle Types and Representative Model						
	JEEP	PICKUP AND UTILITY VAN			WAGON & MINIBUS		
	Toyota Fortuner 2.7 V	Toyota Hilux 4x2 Standard	Toyota Hilux 4x4 Double	Hyundai Sahazor	Toyota Hiace Standard	Toyota Coaster	Hino FB2WGKZ
	CBU	CKD	CBU	CKD	CBU	CBU	CKD
BASIC SPECIFICATIONS							
Engine (cc)	2694	2446	2779	2600	2800	3661	4009
Gross Vehicle Weight (kg)	2800	2580	2590	2734	2850	5180	6200
Number of Axles	2	2	2	2	2	2	2
Types of Tyres	265-65-R17	185-65-R15	205-65-R16	265-65-R17	185-65-R16	205-65-R16	265-65-R17
Seats	7	2	5	3	16	26	26
USAGE							
Life (years)	12	12	12	12	10	10	10
Hours Driven per year	600	650	650	650	1200	1200	1200
Average Speed (km/hr)	60	45	45	45	45	45	45
Km Driven per year	36000	29250	29250	29250	54000	54000	54000
COST (Rs)							
Vehicle Market Price	9,399,000	2,876,500	4,127,000	1,889,000	7,399,000	14,809,000	5,900,000
Market Price of Tyre	22,000	15,800	18,000	21,000	17,000	18,000	21,000
Market Price of Tyre in vehicle	88,000	63,200	72,000	84,000	68,000	72,000	84,000
-	-	-	-	-	-	-	-
Duty / Tax Ratio	0.43	0.27	0.43	0.27	0.43	0.43	0.27
Economic Cost Ratio	0.57	0.73	0.57	0.73	0.57	0.57	0.73
Economic Cost of vehicle	5,357,430	2,099,845	2,352,390	1,378,970	4,217,430	8,441,130	4,307,000
Tyre Cost (Rs)							
Market Price of Tyre	22,000	15,800	18,000	21,000	17,000	18,000	21,000
Tax Ratio on Tyre	0.36	0.36	0.36	0.36	0.36	0.36	0.36
Economic Cost Ratio	0.64	0.64	0.64	0.64	0.64	0.64	0.64
Economic Cost per Tyre Set	14,080	10,112	11,520	13,440	10,880	11,520	13,440

Table – 6 Market / Economic Costs of Bus & Trucks

DESCRIPTION	Vehicle Types and Representative Model				
	Bus	Truck 2 - Axle	Truck 3 - Axle	Truck Articulated	
	Hino AK 1JRA	Hino FL	Hino	Hino SG	Hino FM
	CKD	CKD	CKD	CKD	CKD
BASIC SPECIFICATIONS					
Engine (cc)	7,961	7,961	7,961	8,500	10,520
Gross Vehicle Weight (kg)	14,000	18,000	27,500	35,000	45,000
Number of Axles	2	2	3	4	4
Types of Tyres	1200-20-20R	900 - 20-20R	900 - 20-20R	900 - 20-20R	900 - 20-20R
Seats	54	2	2	3	3
USAGE					
Life (years)	10	10	12	12	12
Hours Driven per year	1,200	1,750	1,750	1,750	1,750
Average Speed (km/hr.)	50	40	40	40	40
Km Driven per year	60,000	70,000	70,000	70,000	70,000
COST (Rs)					
Vehicle Market Price, chassis	11,800,000	10,511,000	13,500,000	12,700,000	15,900,000
Market Price of Tyre	36,500	36,500	36,500	36,500	36,500
Number of tyres	6	6	10	14	14
Market Price of Tyre in vehicle	219,000	219,000	365,000	511,000	511,000
Vehicle Market Price without Tyres	11,800,000	10,511,000	13,500,000	12,700,000	15,900,000
Duty / Tax Ratio	0.27	0.29	0.29	0.29	0.29
Economic Cost Ratio	0.73	0.71	0.71	0.71	0.71
Economic Cost of vehicle	8,614,000	7,462,810	9,585,000	9,017,000	11,289,000
Financial Cost of body	3,186,000	2,837,970	3,645,000	3,429,000	4,293,000
Duty / Tax Ratio	0.13	0.13	0.13	0.13	0.13
Economic Cost Ratio	0.87	0.87	0.87	0.87	0.87
Economic Cost of Body	1,348,500	870,000	1,131,000	1,305,000	1,305,000
Economic Cost of Vehicle	8,614,000	7,462,810	9,585,000	9,017,000	11,289,000
Financial Cost of Vehicle	11,800,000	10,511,000	13,500,000	12,700,000	15,900,000
Tyre Cost (Rs)					
Market Price of Tyre	36,500	36,500	36,500	36,500	36,500
Tax Ratio on Tyre	0.36	0.36	0.36	0.36	0.36
Economic Cost Ratio	0.64	0.64	0.64	0.64	0.64
Economic Cost per Tyre Set	23,360	23,360	23,360	23,360	23,360

2. RESULTS OF ECONOMIC ANALYSIS

There are more than 65 factors that affect the computation of Free Flow Vehicle Operating Cost of a particular vehicle type. However, the riding quality or roughness is the most sensitive parameter in calculation of VOC. Tables 7 to 9 present the VOC (In Economic Terms) of different vehicle types at varying International Roughness Index (IRI) and the Horizontal Curvature (degree/km).

Table – 7

Total Vehicle Operating Cost (Economic) Per 1000 vehicle-km in Rupees

CURVETURE	IRI	Small	Medium	Large	Jeep	Pickup	Wagon	Bus	2XL	3XL	Articulated
deg/km	m/km	Car	Car	Car					Truck	Truck	Truck
0	2	29198.6	29945.0	37564.6	98594.0	92117.1	122210.3	156566.4	146214.5	178484.3	246319.4
0	3	30112.5	31030.7	39207.5	105220.6	95967.6	129026.1	158555.1	154179.5	185800.9	254446.2
0	4	31135.3	32243.0	41095.7	108111.1	100524.5	136129.0	160691.7	162019.3	193041.3	262766.2
0	5	32289.0	33604.7	43277.5	115815.6	105955.1	143666.4	163056.0	169784.6	200241.2	271591.0
0	6	33604.2	35154.1	45819.0	125070.0	112448.8	151786.3	165756.4	177583.3	207478.8	281228.6
0	7	35121.3	36950.3	48808.9	136142.2	120214.1	160605.3	168916.6	185540.6	214846.3	291780.2
0	8	36890.8	39069.4	52354.8	149325.9	129476.7	170178.4	172646.9	193754.2	222416.1	303183.4
0	9	38970.6	41596.9	56578.5	164947.4	140479.2	180489.2	177016.0	202277.8	230225.2	315317.7
0	10	41317.2	44480.3	61370.1	182408.1	152942.7	191467.7	182042.4	211125.1	238281.3	328056.8
0	11	43721.0	47439.0	66240.0	199662.5	165740.2	203017.2	187702.9	220286.2	246573.8	341284.5
0	12	46173.2	50461.5	71163.6	216589.9	178792.0	215035.8	193948.7	229737.7	255084.8	354901.5
0	13	48671.4	53543.1	76133.9	233154.3	192061.3	227431.3	200720.4	239451.1	263794.9	368827.6
0	14	51210.4	56676.6	81142.8	249323.5	205512.2	240124.8	207956.4	249397.4	272686.5	383000.4
0	15	53784.9	60161.6	86629.7	265069.9	219112.6	253052.5	215597.5	259548.4	281744.2	397372.0

Table – 8

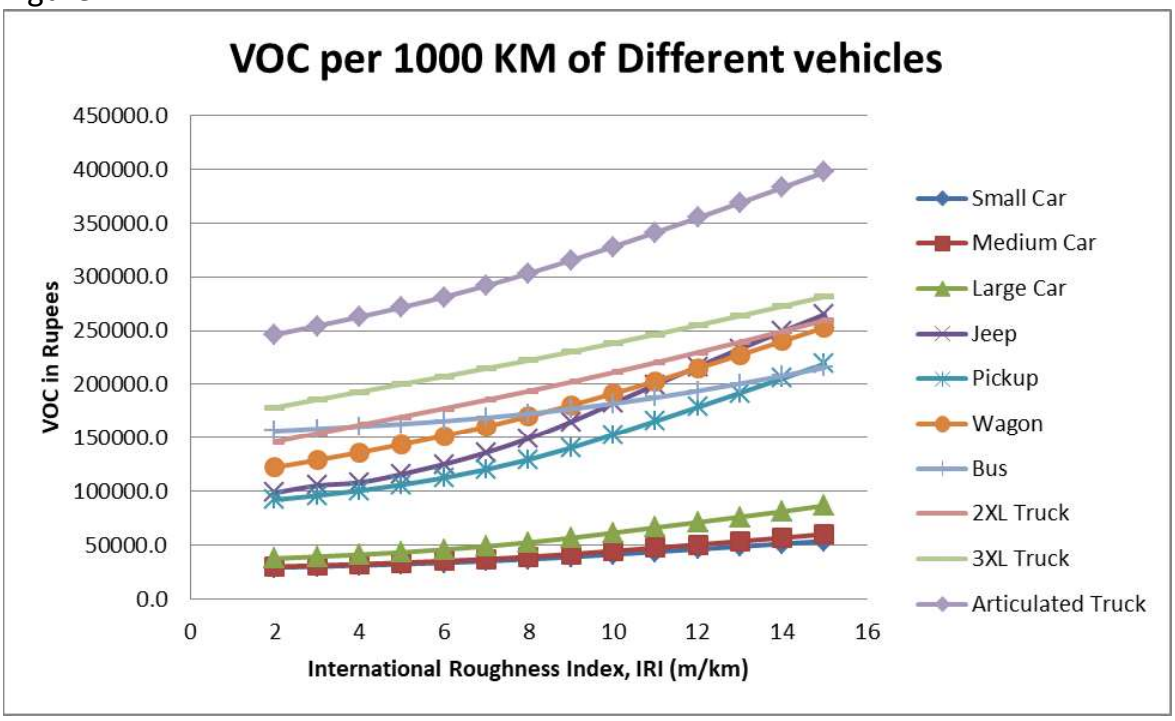
Total Vehicle Operating Cost (Economic) Per 1000 vehicle-km in Rupees

CURVETURE	IRI	Small	Medium	Large	Jeep	Pickup	Wagon	Bus	2XL	3XL	Articulated
deg/km	m/km	Car	Car	Car					Truck	Truck	Truck
150	2	29142.1	29578.7	37410.2	149595.6	104966.0	147399.7	159414.2	146841.8	179876.2	249981.1
150	3	30062.4	30677.3	39058.7	159388.8	109239.0	155435.8	161465.0	154981.0	187398.2	258282.0
150	4	31096.7	31917.8	40964.6	171076.1	114290.4	163813.7	163681.4	163081.5	194941.4	266850.0
150	5	32268.7	33327.0	43179.4	185060.2	120300.4	172690.9	166141.2	171172.1	202521.2	275889.4
150	6	33607.7	34941.5	45766.9	201802.6	127470.9	182215.1	168937.4	179310.8	210166.5	285615.7
150	7	35151.8	36811.3	48807.7	221830.6	136026.2	192503.0	172172.2	187574.1	217915.0	296156.5
150	8	36946.8	39000.7	52399.5	245745.8	146213.1	203615.1	175939.0	196034.8	225806.0	307516.3
150	9	39047.2	41586.5	56657.5	274239.6	158303.7	215547.8	180303.6	204745.0	233870.6	319618.3
150	10	41407.3	44511.6	61469.5	306499.8	171995.6	228242.3	185293.7	213731.4	242127.0	332352.7
150	11	43817.7	47496.8	66348.1	339230.3	186046.7	241605.9	190900.0	222998.2	250581.2	345607.2
150	12	46270.8	50533.5	71272.0	372254.7	200378.0	255534.5	197087.2	232535.1	259230.8	359281.1
150	13	48765.7	53620.7	76237.2	405528.9	214956.3	269927.6	203805.2	242323.3	268068.2	373291.0
150	14	51299.4	56773.7	81268.3	439009.9	229746.7	284695.7	210998.2	252340.6	277084.2	387570.8
150	15	53867.4	60332.1	86856.0	472658.3	244716.1	299764.1	218610.5	262563.2	286268.6	402069.6

Table – 9

Total Vehicle Operating Cost (Economic) Per 1000 vehicle-km in Rupees											
CURVETURE	IRI	Small	Medium	Large	Jeep	Pickup	Wagon	Bus	2XL	3XL	Articulated
deg/km	m/km	Car	Car	Car					Truck	Truck	Truck
300	2	29239.8	29390.9	37511.4	154916.5	109178.7	153473.0	164391.1	148513.2	183176.8	256106.0
300	3	30164.5	30496.9	39159.1	164673.9	113431.9	161497.4	166486.2	156778.8	190817.9	264498.5
300	4	31207.1	31755.2	41071.0	176296.2	118446.8	169843.0	168752.6	165063.3	198516.1	273152.7
300	5	32391.9	33195.7	43300.7	190175.6	124396.1	178651.3	171264.4	173390.5	206287.8	282235.0
300	6	33748.7	34855.9	45911.9	206765.5	131474.8	188050.5	174103.4	181794.8	214151.3	291917.2
300	7	35313.3	36781.9	48981.7	226589.9	139902.5	198142.2	177354.0	190319.9	222127.7	302328.9
300	8	37129.0	39030.3	52602.1	250253.0	149924.0	208986.1	181094.0	199012.7	230238.6	313521.1
300	9	39245.7	41669.1	56881.8	278454.6	161814.4	220594.1	185383.8	207913.6	238503.6	325465.4
300	10	41616.1	44635.5	61705.6	310396.2	175279.5	232932.8	190257.5	217050.0	246937.9	338081.0
300	11	44029.2	47648.1	66585.3	342796.4	189087.4	245935.6	195721.4	226434.3	255551.2	351266.9
300	12	46478.6	50699.3	71501.6	375492.5	203169.8	259519.1	201756.5	236066.4	264347.0	364923.4
300	13	48965.0	53790.5	76453.6	408451.6	217503.1	273595.3	208327.1	245936.7	273324.7	378962.1
300	14	51486.3	57169.0	81798.9	441638.0	232059.5	288081.5	215388.0	256030.2	282480.5	393311.7
300	15	54040.6	60685.9	87322.2	475016.3	246810.6	302904.0	222889.0	266328.9	291808.6	407915.3

Figure – 1



3. COMPONENTS OF VOC

Free-Flow Vehicle Operating Costs Model (HDM-VOC) output comprises the following components: -

- Fuel
- Lubricants
- Tires
- Crew Time
- Passenger Time
- Cargo Holding
- Maintenance Labor
- Maintenance Parts
- Depreciation
- Interest
- Overhead

The analysis reveals that on the average fuel contributes about 36% in the VOC of any vehicle. If lubricant and tire consumption are also added then the contribution is about 47%. It means that if the VOC of any vehicle is Rupee one per kilometer than Fuel, Lubricant and Tires consume nearly 47 paisa.

Figure – 2 present the distribution of components of VOC of different vehicle types.

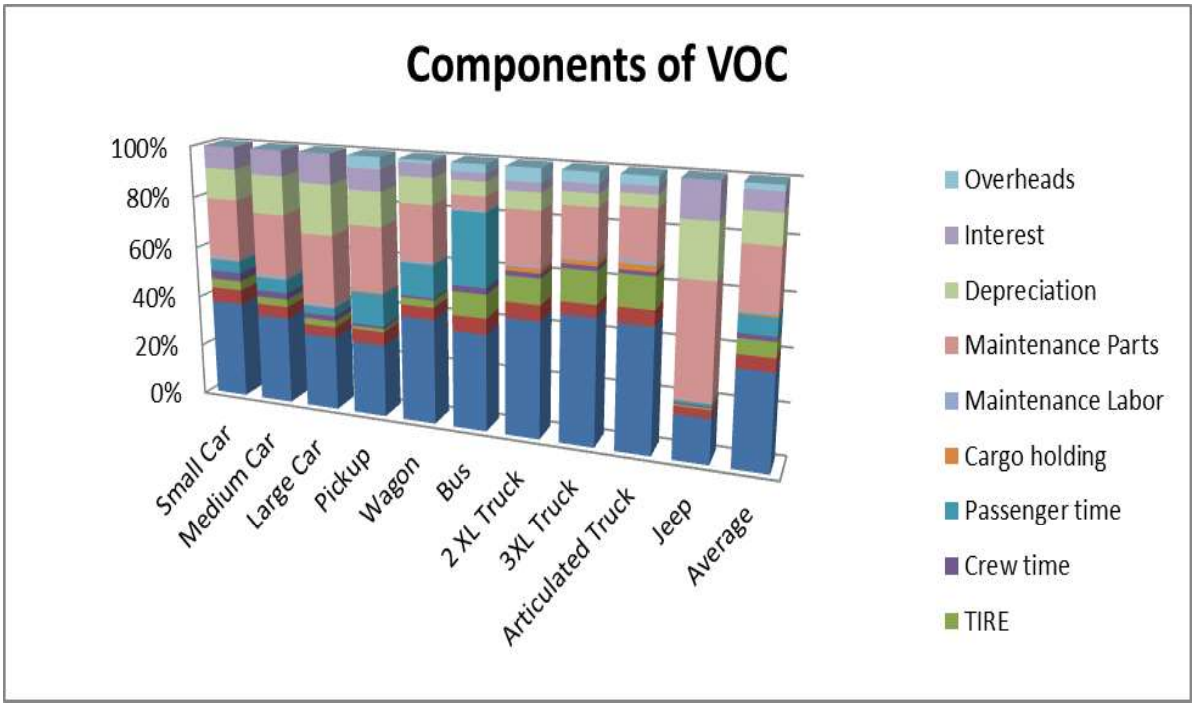


Table- 10 shows percentage contribution of Fuel, Lubricant and tires in VOC of different types of vehicle. The contribution of fuel in the total VOC for different vehicles is ranged from 17% to 45%.

Table – 10 Percentage Distributions of VOC Components

VEHICLE	Fuel	Lubricants	TIRE	Crew time	Passenger time	Cargo holding	Maintenance Labor	Maintenance Parts	Depreciation	Interest	Overheads	TOTAL
	%	%	%	%	%	%	%	%	%	%	%	%
Small Car	37.7	5.7	3.6	3.3	4.6	0.3	1	23.1	12	8.7	0	100
Medium Car	34.1	4.7	3	2.5	5.3	0.1	0.9	24.4	15.1	9.9	0	100
Large Car	28.9	4	2.5	2.1	3	0.2	0.7	27.4	19.6	11.6	0	100
Pickup	28.2	5.3	1.2	1	12.4	0.5	0.3	25	13.3	8.3	4.5	100
Wagon	40.6	4.7	2.8	0.8	12.6	0.1	0.6	21.6	9.9	5.2	1.1	100
Bus	37.4	6.1	9	2.4	27.7	0.1	0.7	5	5.4	3	3.2	100
2 XL Truck	44.8	6	9.8	1.8	0	1.8	0.7	20.1	6.4	3.4	5.2	100
3XL Truck	48.4	4.8	12.1	1.7	0	1.7	0.7	18.4	4.8	3.3	4.1	100
Articulated Truck	47.6	5.9	12	1.4	0	2.3	1.2	18.7	4.6	3	3.3	100
Jeep	17	3.2	0.7	0.6	0.8	0.1	0.2	43.2	20.5	13.7	0	100
Average	36.47	5.04	5.67	1.76	6.64	0.72	0.7	22.69	11.16	7.01	2.14	100